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B. Tech MID-SEMESTER EXAMINATION, September 2024

5th Semester

Course Code-. CICPC15

Course Title-. Artificial Intelligence & Machine Learning

Time- 1.5 Hour

Max. Marks- 15.

Note: Attempt all questions. Missing data/information (if any) may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	What is the Turing Test, and how does it relate to the concept of Artificial Intelligence (AI)?	1	CO1
1b	Explain the concept of "bounded rationality" in the context of AI, and provide a real-world example where it might be relevant.	2	CO1
2a	Explain 8-puzzle problem	1	CO1
2b	Compare and Contrast breadth-first search (BFS) and Depth-first search (DFS), highlighting their strengths and weaknesses in different problem-solving scenarios.	2	CO1
3a	Whether a search is guaranteed to find an optimal solution if a suitable heuristic is used?	1	CO2
3b	What is the main difference between an informed search algorithm and an uninformed search algorithm? Provide an example of each	2	CO2
4a	Explain N-Queens Problem.	1	CO2
4b	Describe the steps involved in solving a Constraint Satisfaction Problem (CSP) using a backtrack search.	2	CO2
5a	In-game Theory, what is a Nash Equilibrium?	1	CO2
5b	Compare and contrast local search algorithms with systematic search algorithms providing examples of when each might be appropriate	2	CO2